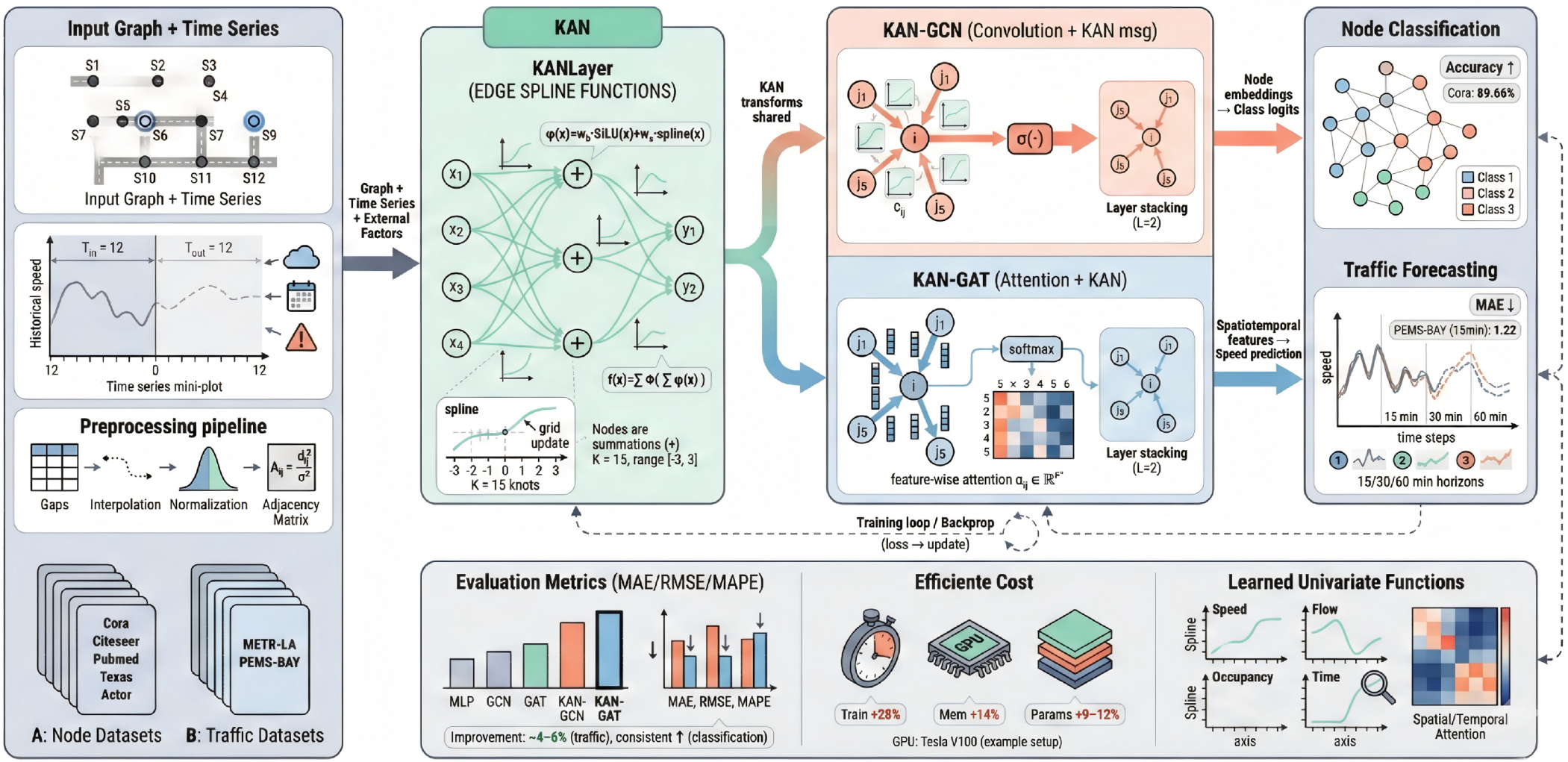
Input Graph + Time Series
S1
S2
S3
S4
S5
S6
S7
S7
S9
S10
S11
S12
Input Graph + Time Series
T_{in} = 12
T_{out} = 12
Historical speed
12
0
12
Time series mini-plot
Preprocessing pipeline
Gaps
Interpolation
Normalization
$A_{ij} = \frac{d_{ij}^2}{\sigma^2}$
Adjacency Matrix
Cora
Citeseer
Pubmed
Texas
Actor
METR-LA
PEMS-BAY
A: Node Datasets
B: Traffic Datasets
Graph + Time Series + External Factors
KAN
KANLayer
(EDGE SPLINE FUNCTIONS)
$\varphi(x)=w_b \cdot SiLU(x)+w_s \cdot spline(x)$
x_1
x_2
x_3
x_4
y_1
y_2
$f(x)=\sum \Phi(\sum \varphi(x))$
spline
grid update
-3 -2 -1 0 1 2 3
K = 15 knots
Nodes are summations (+)
K = 15, range [-3, 3]
KAN transforms shared
KAN-GCN (Convolution + KAN msg)
j_1
j_5
i
c_{ij}
σ(·)
Layer stacking
(L=2)
KAN-GAT (Attention + KAN)
softmax
5 × 3 4 5 6
feature-wise attention $\alpha_{ij} \in \mathbb{R}^{F'}$
Layer stacking
(L=2)
Node embeddings → Class logits
Spatiotemporal features → Speed prediction
Node Classification
Accuracy ↑
Cora: 89.66%
Class 1
Class 2
Class 3
Traffic Forecasting
MAE ↓
PEMS-BAY (15min): 1.22
speed
15 min
30 min
60 min
time steps
15/30/60 min horizons
Training loop / Backprop
(loss → update)
Evaluation Metrics (MAE/RMSE/MAPE)
MLP
GCN
GAT
KAN-GCN
KAN-GAT
MAE, RMSE, MAPE
Improvement: ~4–6% (traffic), consistent ↑ (classification)
Efficiente Cost
GPU
Train +28%
Mem +14%
Params +9–12%
GPU: Tesla V100 (example setup)
Learned Univariate Functions
Speed
Flow
Occupancy
Time
Spline
axis
Spatial/Temporal Attention